

# Distance-adaptive unsupervised CNN model for computer-generated holography

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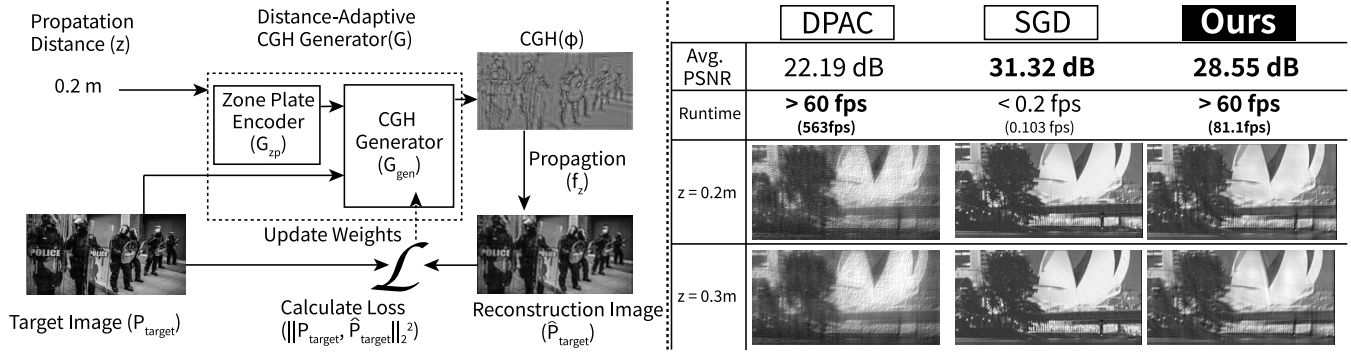


Figure 1: Left: Our model can receive propagation distance and target image, then generates CGH. Right: Reconstruction images of CGH generated by DPAC, SGD, ours.

## ABSTRACT

Convolutional neural networks (CNN) are useful for overcoming the trade-off between generation speed and accuracy in the process of synthesizing computer-generated holograms (CGH). However, methods using a CNN cannot specify the propagation distance when reproducing a hologram, thereby limiting their practical usage across various contexts. Therefore, in this study, we developed a CNN model that can generate CGH by specifying the target image and propagation distance. The proposed method demonstrates performance comparable to traditional methods with a fixed distance and achieves the generation accuracy and speed necessary for practical use even when the propagation distance is changed.

## CCS CONCEPTS

• Human-centered computing → Displays and imagers.

## KEYWORDS

hologram, optimization, neural network

## ACM Reference Format:

Yuto Asano, Kenta Yamamoto, Tatsuki Fushimi, and Yoichi Ochiai. 2024. Distance-adaptive unsupervised CNN model for computer-generated holography. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641234.3671051>

## 1 INTRODUCTION

Creating computer-generated holography (CGH) rapidly and accurately is crucial for technologies like holographic near-eye displays and holographic projectors. Recently, convolutional neural networks (CNNs) have become a popular method to improve CGH generation. For instance, Horisaki *et al.* used a basic U-Net model to create CGH, while Peng *et al.* developed a specialized model which can generate CGH with a peak signal-to-noise ratio (PSNR) greater than 30 dB and also reach or exceed 30 fps.

However, there's still a challenge in producing CGH that works well across various distances without needing retraining for each new distance. Current CNN models can manage this trade-off at a fixed distance but struggle with multiple distances in real-time.

To solve this problem, we introduce a new CNN-based approach that adapts to different distances. Our model takes both the target image and the desired propagation distance as inputs, enabling it to generate precise CGHs suitable for any specified distance. Our

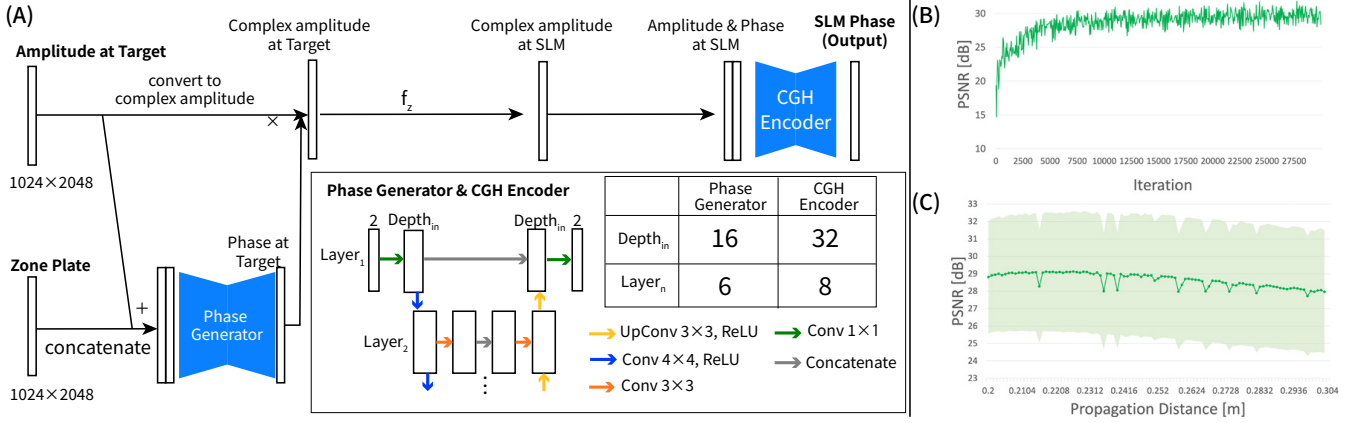
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SIGGRAPH Posters '24, July 27–August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0516-8/24/07

<https://doi.org/10.1145/3641234.3671051>



**Figure 2: (A) Distance-adaptive CGH generator architecture. Our model consists of a phase generator and a CGH encoder. (B) Training result of our model. (C) Evaluated accuracy of the CGH output by our model for each propagation distance. The background colors represent the standard deviations.**

model successfully generates accurate CGHs across more than 100 different distances, covering a range of over 0.1 meters.

## 2 METHOD

Our goal is to develop a distance-adaptive CGH generator  $G$  which receives both target image  $P_{target}$  and propagation distance  $z$ , and outputs an appropriate CGH  $\phi$  (Figure 1 (Left)). When a coherent, uniform-amplitude light field is incident on a phase-only spatial light modulator (SLM), the light field is modulated by  $\phi(x, y)$ . The modulated light field propagates to a screen which is located at a distance  $z$  from the SLM, according to the propagation function  $f_z$ . We employed angular spectrum method [Goodman 2005] as  $f_z$ . Reconstruction image  $\hat{P}_{target}$  on the screen is denoted as  $\hat{P}_{target} = |f_z(\phi)|$ . Our goal can be represented as a solving optimization problem of the form

$$\min_G \mathcal{L}(G) = \|P_{target} - \hat{P}_{target}\|_2^2 \quad (1)$$

where  $\mathcal{L}$  is a loss function and we adopted mean squared error.

To solve this problem, we designed  $G$  from two components: zone plate encoder  $G_{zp}$  and CGH generator  $G_{gen}$ . A zone plate is a well-known CGH for point light sources. The zone plate encoder receives propagation distance and encodes zone plate according to  $G_{zp}(z) = \arg(f_z(P_{pl}))$ , where  $P_{pl}$  denotes the amplitude distribution of the point light source and  $\arg(x)$  represents the angle of the complex number  $x$ . The value of the central point of  $P_{pl}$  is 1, and all other regions are 0. The CGH generator is a neural network which accepts both zone plate and target image, and generates CGH. The architecture of this CNN is based on HoloNet [Peng et al. 2020] (Figure 2 (Left)). Using  $G_{zp}$  and  $G_{gen}$ ,  $\hat{P}_{target}$  can be represented as

$$\begin{aligned} \hat{P}_{target} &= |f_z(\phi)| \\ &= |f_z(G(P_{target}, z))| \\ &= |f_z(G_{gen}(P_{target}, G_{zp}(z)))| \end{aligned} \quad (2)$$

The training of  $G_{gen}$  is a task that updates the weights of  $G_{gen}$  to solve Equation 1. We used 3,450 images from the DIV2K [Agustsson and Timofte 2017] and the Flickr2K for training, 100 images from

the DIV2K validation datasets for validation. We executed 30,000 training iterations and validated peak signal-to-noise ratio (PSNR) after every 50 iterations. All training and evaluation processes were executed using a Tesla V100 SXM2 graphics processing unit, Python 3.7.13, and PyTorch 1.13.1. Training process took about 6 hours under this condition.

## 3 RESULT

During training, our model stabilizes near a PSNR of 30 after 15,000 iterations (Figure 2 (B)). To evaluate the model's performance, we recorded the average PSNR of 100 CGH images outputted by the model at 100 different propagation distances. Our model achieved high precision at all distances (Figure 2 (C)). We compared our model to existing methods, double phase-amplitude coding (DPAC) [Maimone et al. 2017] and stochastic gradient descent (SGD) [Peng et al. 2020]. The results are presented in the Figure 1 (Right). Our model better accuracy than DPAC, and close to 30 dB. The frames per second (fps) exceeded 60, a value that is satisfactorily sufficient for display purposes.

## 4 CONCLUSION

In this study, we developed a CNN that outputs high-precision CGH despite specifying the propagation distance as an input. A future challenge is to develop a general-purpose CNN model that can adapt to changes in other optical calculation parameters, such as wavelength and pitch size, as well as to human eye aberrations.

## REFERENCES

- Eirikur Agustsson and Radu Timofte. 2017. NTIRE 2017 Challenge on Single Image Super-Resolution: Dataset and Study. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*.
- Joseph W. Goodman. 2005. *Introduction to Fourier optics* (3rd ed. ed.). Roberts, Englewood.
- Andrew Maimone, Andreas Georgiou, and Joel S. Kollin. 2017. Holographic Near-Eye Displays for Virtual and Augmented Reality. *ACM Trans. Graph.* 36, 4, Article 85 (jul 2017), 16 pages. <https://doi.org/10.1145/3072959.3073624>
- Yifan Peng, Suyeon Choi, Nitish Padmanaban, and Gordon Wetzstein. 2020. Neural Holography with Camera-in-the-Loop Training. *ACM Trans. Graph.* 39, 6, Article 185 (nov 2020), 14 pages. <https://doi.org/10.1145/3414685.3417802>